

ECON2043 Experimental and Behavioural Economics

Seminar 3: Problem Set (Risk Preferences)

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Activity 1

Part 1. Identify

- ▶ Research question(s)
- ▶ How do they solve it: method, data, and experimental design
- ▶ What do they find?

Part 2. Evaluate the big picture

- ▶ Is the research question interesting?
- ▶ Is it important?
- ▶ Is it novel?
- ▶ Is it non-obvious?

Activity 2

A good research question should be:

- ▶ specific and clear
- ▶ focused
- ▶ testable
- ▶ feasible

Example

Does peer acceptance affect the acceptability of AI diagnostics?

Experimental Design

Treatment Group 1

- ▶ Participants are told that 80% of peers accept AI consultations.

Treatment Group 2¹

- ▶ Participants are told that 80% of peers reject AI consultations.

Control Group

- ▶ No information about peer opinions is provided.

Choice and Trust Assessment

- ▶ Participants make their selection and report confidence on a 7-point Likert scale.

Survey

- ▶ Collect demographics such as age, gender, medical knowledge, and perceptions about AI.

¹Given the module size, one treatment group is usually cleaner.

Question 1

Suppose a decision maker can choose between a gamble that pays 1 with probability $\frac{2}{3}$ and 0 otherwise, or a sure payment of 0.5.

- (a) What would a risk averse decision maker with utility $u(x) = \sqrt{x}$ choose? What would a risk neutral decision maker choose?
- (b) What would a risk loving decision maker with utility $u(x) = x^2$ choose?

Answer 1

Expected value benchmark

$$\mathbb{E}[X] = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{2}{3} > 0.5$$

So a risk neutral decision maker chooses the gamble.

Risk averse, $u(x) = \sqrt{x}$

$$U(\text{gamble}) = \frac{2}{3}\sqrt{1} + \frac{1}{3}\sqrt{0} = \frac{2}{3}$$

$$U(\text{sure}) = \sqrt{0.5} \approx 0.707$$

Hence the risk averse decision maker prefers the sure payment.

Risk loving, $u(x) = x^2$

$$U(\text{gamble}) = \frac{2}{3} \cdot 1^2 + \frac{1}{3} \cdot 0^2 = \frac{2}{3}$$

$$U(\text{sure}) = (0.5)^2 = 0.25$$

Hence the risk loving decision maker prefers the gamble.

Question 2

Your main job today is to read Holt and Laury (2002). Please take it seriously. You want to understand the four key questions covered in the lecture. The paper mentions two common classes of risk preferences, CRRA and CARA.

(a) For CRRA preferences,

$$u(x) = \frac{x^{1-r}}{1-r}$$

show that rescaling payoffs by a factor $k > 0$ does not affect preferences over gambles. Start with $r = \frac{1}{2}$ and compare:

- ▶ gamble A : payoff 4 with probability $\frac{1}{2}$ and 0 otherwise
- ▶ riskless outcome B : payoff 1 for sure

What does the decision maker prefer for $k = 1$, for $k = 10$, and for general k ? Then show the result formally for any two gambles.

(b) For CARA preferences,

$$u(x) = -e^{-\alpha x}$$

show formally that the initial wealth level w does not affect preferences over any two gambles when the relevant payoff is $w + X$.

Answer 2(a): Example and Intuition

For $r = \frac{1}{2}$, the CRRA utility is proportional to $\sqrt{x}$, so it is enough to compare expected values of $\sqrt{x}$.

Case $k = 1$

$$\mathbb{E}[u(A)] = \frac{1}{2}\sqrt{4} + \frac{1}{2}\sqrt{0} = 1$$

$$\mathbb{E}[u(B)] = \sqrt{1} = 1$$

The decision maker is indifferent.

Case $k = 10$

$$\mathbb{E}[u(10A)] = \frac{1}{2}\sqrt{40} = \sqrt{10}$$

$$\mathbb{E}[u(10B)] = \sqrt{10}$$

Again the decision maker is indifferent.

General $k > 0$

$$\mathbb{E}[u(kA)] = \frac{1}{2}\sqrt{4k} = \sqrt{k} \quad \mathbb{E}[u(kB)] = \sqrt{k}$$

So scaling all payoffs by the same positive factor does not change the ranking.

Answer 2(a): Formal CRRA Proof

Let X be any gamble with outcomes x and probabilities $f(x)$. Under CRRA utility,

$$\mathbb{E}[u(kX)] = \sum_x f(x) \frac{(kx)^{1-r}}{1-r} = k^{1-r} \sum_x f(x) \frac{x^{1-r}}{1-r} = k^{1-r} \mathbb{E}[u(X)]$$

with $k > 0$.

Now compare two gambles A and B .

$$\mathbb{E}[u(kA)] \geq \mathbb{E}[u(kB)] \iff k^{1-r} \mathbb{E}[u(A)] \geq k^{1-r} \mathbb{E}[u(B)]$$

Because $k^{1-r} > 0$, multiplying both sides by this common positive factor does not change the inequality. Therefore,

$$\mathbb{E}[u(A)] \geq \mathbb{E}[u(B)] \iff \mathbb{E}[u(kA)] \geq \mathbb{E}[u(kB)]$$

So CRRA preferences are invariant to proportional rescaling of all payoffs.

Answer 2(b): Formal CARA Proof

Let X be any gamble and w be initial wealth. Under CARA utility,

$$u(x) = -e^{-\alpha x}$$

we have

$$\mathbb{E}[u(w + X)] = \sum_x f(x) \left(-e^{-\alpha(w+x)} \right) = e^{-\alpha w} \sum_x f(x) \left(-e^{-\alpha x} \right) = e^{-\alpha w} \mathbb{E}[u(X)]$$

Now compare two gambles A and B :

$$\mathbb{E}[u(w + A)] \geq \mathbb{E}[u(w + B)] \iff e^{-\alpha w} \mathbb{E}[u(A)] \geq e^{-\alpha w} \mathbb{E}[u(B)]$$

Since $e^{-\alpha w} > 0$, the common factor cancels. Therefore,

$$\mathbb{E}[u(A)] \geq \mathbb{E}[u(B)] \iff \mathbb{E}[u(w + A)] \geq \mathbb{E}[u(w + B)]$$

So under CARA preferences, the starting wealth level does not affect the ranking of gambles.

Question 3

For this question, let random variable X have finite support $\Omega = \{x_1, x_2, \dots, x_N\}$ and probability mass function $f(x) = \Pr(X = x)$. Recall:

$$\mathbb{E}[X] = \sum_{x \in \Omega} x f(x)$$

(a) Compute $\mathbb{E}[X]$ when outcomes are 1, 2, 3, 4 with equal probability.

(b) Show that for constants a and b ,

$$\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$$

(c) For constants a , b , c and random variables X , Y , show that

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

(d) Compute $\mathbb{E}[X]$ for outcomes 0, 2, 4 with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$. Compute $\mathbb{E}[Y]$ for outcomes 0, 4, 16 with the same probabilities.

(e) State the general formula for $\mathbb{E}[g(X)]$. Check it for $g(X) = X^2$ using part (d).

(f) Let $X \in \{0, 1, 2\}$ be equally likely. Compute $\mathbb{E}[\sqrt{X}]$, $\mathbb{E}[X]$, and $\mathbb{E}[X^2]$. Compare each with $g(\mathbb{E}[X])$.

(g) Optional: what general principle does part (f) illustrate?

Answer 3(a)-(c)

(a)

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

(b)

$$\mathbb{E}[a + bX] = \sum_x (a + bx)f(x) = a \sum_x f(x) + b \sum_x xf(x) = a + b\mathbb{E}[X]$$

(c) Let $f_{X,Y}(x, y)$ denote the joint probability mass function of (X, Y) . Then

$$\begin{aligned}\mathbb{E}[aX + bY + c] &= \sum_x \sum_y (ax + by + c)f_{X,Y}(x, y) \\ &= a \sum_x \sum_y xf_{X,Y}(x, y) + b \sum_x \sum_y yf_{X,Y}(x, y) + c \sum_x \sum_y f_{X,Y}(x, y) \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y] + c\end{aligned}$$

This is the linearity of expectation.

Answer 3(d)-(e)

(d)

$$\mathbb{E}[X] = 0 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} + 4 \cdot \frac{1}{2} = 2.5$$

$$\mathbb{E}[Y] = 0 \cdot \frac{1}{4} + 4 \cdot \frac{1}{4} + 16 \cdot \frac{1}{2} = 9$$

(e) For any function g ,

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega} g(x)f(x)$$

If $g(X) = X^2$ in part (d), then

$$\mathbb{E}[g(X)] = 0^2 \cdot \frac{1}{4} + 2^2 \cdot \frac{1}{4} + 4^2 \cdot \frac{1}{2} = 0 + 1 + 8 = 9$$

which matches the value of $\mathbb{E}[Y]$ because $Y = X^2$ for those outcomes.

Answer 3(f)-(g)

(f) If $X \in \{0, 1, 2\}$ is equally likely, then

$$\mathbb{E}[\sqrt{X}] = \frac{\sqrt{0} + \sqrt{1} + \sqrt{2}}{3} = \frac{1 + \sqrt{2}}{3} \approx 0.80$$

$$\mathbb{E}[X] = \frac{0 + 1 + 2}{3} = 1 \quad \mathbb{E}[X^2] = \frac{0^2 + 1^2 + 2^2}{3} = \frac{5}{3} \approx 1.67$$

Comparisons:

$$\mathbb{E}[\sqrt{X}] < \sqrt{\mathbb{E}[X]} = 1$$

$$\mathbb{E}[X] = \mathbb{E}[X]$$

$$\mathbb{E}[X^2] > (\mathbb{E}[X])^2 = 1$$

(g) This is Jensen's inequality:

► if h is concave, then $\mathbb{E}[h(X)] \leq h(\mathbb{E}[X])$

► if g is convex, then $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$

Here, $\sqrt{x}$ is concave, x is linear, and x^2 is convex.